

MOBILE & WIRELESS COMMUNICATION

Lecture 4

Orthogonal Frequency Division Multiplexing

Module 1: Wireless Channels

Following A. K. Jagannatham, Principles of Modern Wireless Communication Systems (McGraw Hill, 2016), Chapter 7 in full, Sections 7.1 to 7.8.

**ISI problem → orthogonal subcarriers → IFFT/FFT → cyclic prefix
BER, MIMO-OFDM, carrier frequency offset, PAPR, and SC-FDMA**

Learning Objectives

- Explain why a wideband single-carrier signal suffers ISI over a frequency-selective channel.
- Derive the multicarrier model and prove subcarrier orthogonality by direct integration.
- Show the sampled multicarrier signal IS an IDFT, enabling IFFT/FFT implementation.
- Derive the cyclic prefix, circular convolution, and the per-subcarrier model $Y(k) = H(k)X(k) + N(k)$.
- Derive the OFDM BER and compute practical system parameters for WiMAX.
- Extend OFDM to MIMO-OFDM across subcarriers.
- Derive the ICI caused by carrier frequency offset and the resulting SINR.
- Derive the PAPR of OFDM and explain how SC-FDMA eliminates it.

PART STRUCTURE



Motivation



Multicarrier & Orthogonality



IFFT/FFT



Cyclic Prefix



Performance & MIMO-OFDM



CFO, PAPR, SC-FDMA

The Problem: ISI over a Frequency-Selective Channel

From Lecture 2: when the signal bandwidth exceeds the coherence bandwidth, the channel is frequency-selective and each received sample is corrupted by previous symbols:

$$y(n) = h(0)x(n) + h(1)x(n-1) + \dots + h(L-1)x(n-L+1)$$

ISI component

For a single-carrier system the symbol time and rate are set directly by the bandwidth:

$$T = \frac{1}{B}, \quad \text{Rate} = \frac{1}{1/B} = B$$

HIGHER RATE

demands larger bandwidth B , which shortens the symbol time $T = 1/B$

SHORTER SYMBOLS

become comparable to the channel delay spread, so ISI worsens

EQUALIZATION

complexity grows with delay spread and becomes impractical

The OFDM idea: rather than fight the frequency-selective channel, split the band so that no piece of the signal ever sees it.

Dividing the Band into N Subcarriers

Divide the total bandwidth B into N sub-bands of width B/N . The i th subcarrier sits at:

$$f_i = i \frac{B}{N}, \quad -(N/2 - 1) \leq i \leq N/2$$

Modulating symbol X_i onto subcarrier i and superposing all N gives the composite transmit signal:

$$s_i(t) = X_i e^{j2\pi f_i t} = X_i e^{j2\pi i \frac{B}{N} t}$$

$$s(t) = \sum_i s_i(t) = \sum_i X_i e^{j2\pi i \frac{B}{N} t}$$

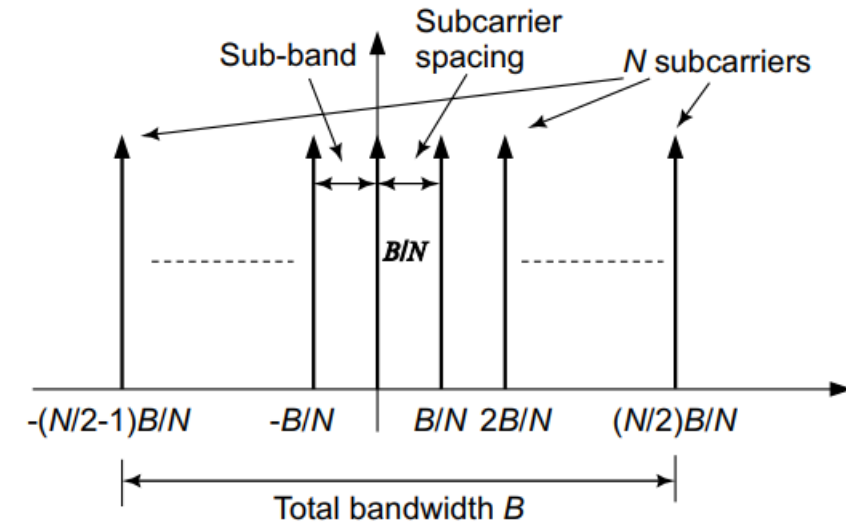


Figure 7.1 Multi-carrier concept

This is exactly a Fourier series with fundamental frequency $f_0 = B/N$, where the data symbols are the coefficients.

Proving Subcarrier Orthogonality

To extract X_l , correlate the received signal with the l th subcarrier over one symbol period $T_0 = N/B$:

$$f_0 \int_0^{T_0} y(t) (e^{j2\pi f_l t})^* dt = \frac{B}{N} \int_0^{N/B} \sum_i X_i e^{j2\pi i \frac{B}{N} t} e^{-j2\pi l \frac{B}{N} t} dt$$

Split the sum into the wanted term and the rest:

$$= \frac{B}{N} \int_0^{N/B} \underset{i=l}{X_l} dt + \frac{B}{N} \sum_{i \neq l} X_i \int_0^{N/B} \underset{=0}{e^{j2\pi(i-l)f_0 t}} dt = X_l$$

The interference vanishes because of the orthogonality of the complex exponentials:

$$\int_0^{N/B} e^{j2\pi(i-l)\frac{B}{N}t} dt = \begin{cases} 0, & i \neq l \\ \frac{N}{B}, & i = l \end{cases}$$

WHY THE INTEGRAL IS ZERO

For $i \neq l$ the integrand is a sinusoid at frequency $(i-l)f_0$, an INTEGER multiple of the fundamental. Integrating a whole number of complete cycles over the period gives exactly zero. This is the 'orthogonal' in OFDM.

Same Rate, No ISI: The Payoff

Multicarrier sends N symbols in parallel over a window N/B , so the overall rate is unchanged:

$$\text{Symbol rate} = \frac{N}{N/B} = B \quad (\text{unchanged from single carrier})$$

What is gained is that each subcarrier is now narrowband:

$$\frac{B}{N} \ll B_c \Rightarrow \text{each subcarrier sees flat fading}$$

$$B = 1.024 \text{ MHz}, N = 256 \Rightarrow B_s = \frac{1024}{256} = 4 \text{ kHz} \ll B_c \approx 250 \text{ kHz}$$

The wideband 1.024 MHz signal was firmly frequency-selective against a 250 kHz coherence bandwidth. Every 4 kHz subcarrier is comfortably flat.

Same data rate, no ISI, no equalizer. That is the entire payoff of multicarrier modulation.

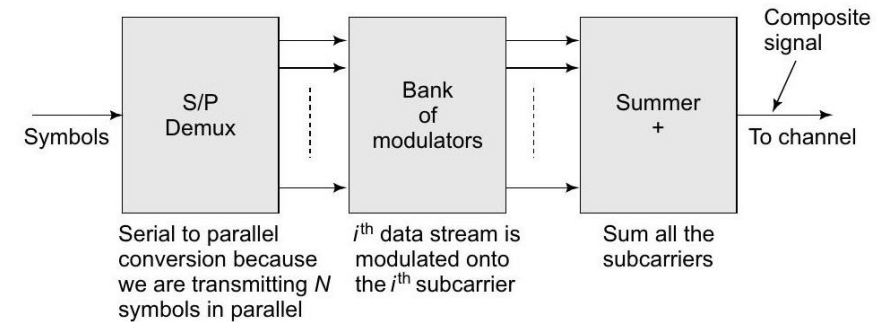


Figure 7.2 Multicarrier modulation transmitter

Figure 7.2: Multicarrier modulation transmitter

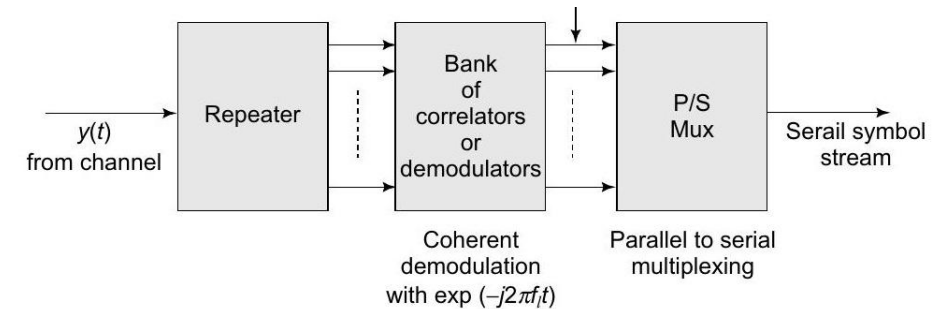


Figure 7.3 Multi-carrier modulation receiver

Figure 7.3: Multicarrier modulation receiver (Jagannatham, 2016)

The Implementation Bottleneck

- The multicarrier scheme just derived needs a bank of N separate modulators at the transmitter and N correlators at the receiver.
- With $N = 256$ or 1024 closely spaced subcarrier oscillators, building and stabilizing that many local oscillators is impractical.
- This is the barrier that kept multicarrier modulation out of deployment for decades.

THE 1971 BREAKTHROUGH

Weinstein and Ebert, at Bell Telephone Laboratories, showed the entire modulator bank could be replaced by a single Discrete Fourier Transform. Their paper: 'Data Transmission by Frequency Division Multiplexing using the Discrete Fourier Transform.'

The next slide shows why the DFT appears, and it is almost an accident of sampling.

Sampling Reveals an IDFT

The composite signal is band-limited to B , so the Nyquist sampling interval is $T_s = 1/B$. Sampling $s(t)$ at $t = uT_s$:

$$T_s = \frac{1}{B}, \quad x(u) = s(uT_s) = \sum_i X_i e^{j2\pi i \frac{Bu}{NB}} = \sum_i X_i e^{j2\pi i \frac{u}{N}}$$

The bandwidth B cancels completely, leaving an expression that depends only on the index ratio u/N . Compare with the definition of the inverse DFT:

$$x(u) = \text{IDFT}\{X(0), X(1), \dots, X(N-1)\}$$

THEY ARE THE SAME EXPRESSION

The u th time sample of the multicarrier signal IS the u th IDFT coefficient of the data symbols. So, the transmitter generates the signal with one IFFT instead of N modulators, and the receiver recovers the data with one FFT.

NO OSCILLATORS

One IFFT block replaces the entire bank of N modulators

$O(N \log N)$

The FFT makes large N computationally cheap

ALL DIGITAL

Implemented in baseband DSP, not analogue hardware

OFDM Transmitter and Receiver Structure

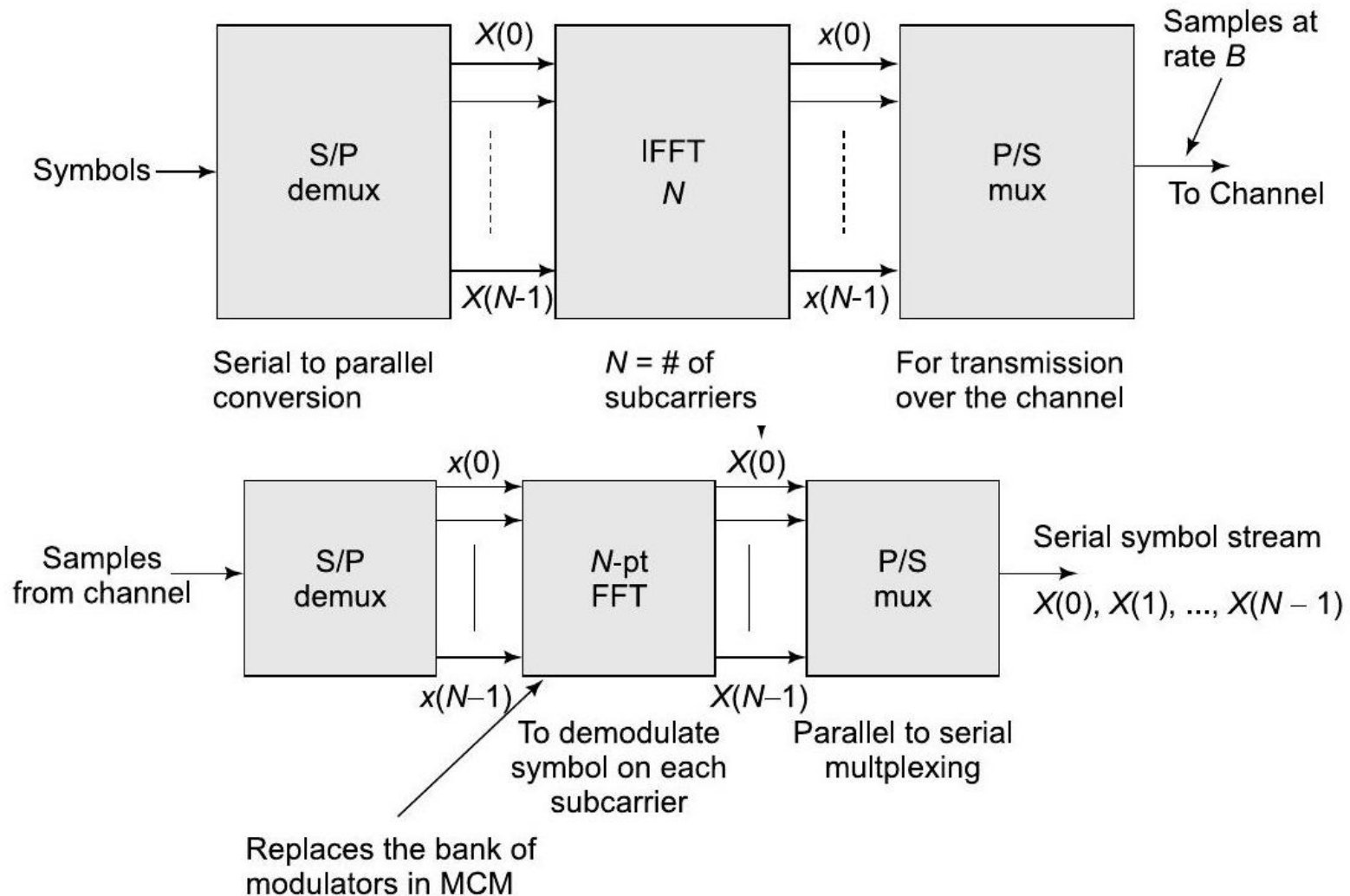


Figure 7.5 *OFDM receiver schematic with FFT*

A Remaining Problem: Inter-OFDM-Symbol Interference

Send two consecutive OFDM blocks. The first sample of the current block still picks up energy from the TAIL of the previous block:

$$y(0) = h(0)x(0) + h(1)\tilde{x}(N-1) + \cdots + h(L-1)\tilde{x}(N-L+1)$$

ISI from PREVIOUS OFDM symbol

where $\tilde{x}$ denotes samples of the PREVIOUS OFDM symbol.

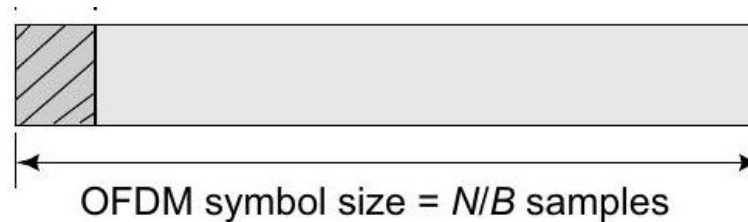


Figure 7.6 *Inter-OFDM symbol interference*

THE IFFT ALONE WAS NOT ENOUGH

The first $L-1$ samples of every OFDM block are corrupted by the previous block, so the clean per-subcarrier separation we derived does not yet hold. One more idea is needed, and it is a surprisingly cheap one.

The Cyclic Prefix

Prepend the LAST L_c samples of the block to its own front, cycling the tail around to the beginning:

$$x(N - L_c), \dots, x(N - 1), x(0), x(1), \dots, x(N - 1)$$

cyclic prefix

current block

Now the same first sample picks up interference from $x(N-1), x(N-2), \dots$ which belong to the SAME symbol:

$$y(0) = h(0)x(0) + h(1)x(N - 1) + \dots + h(L - 1)x(N - L + 1)$$

ISI from the SAME OFDM symbol

provided the prefix is at least as long as the channel memory:

$$L_c \geq L - 1$$

- Inter-OFDM-symbol interference is eliminated entirely: no energy from the previous block reaches the current one.
- The prefix is DISCARDED at the receiver before the FFT, so it carries no information and costs only rate.

Why Cyclic: Linear Becomes Circular Convolution

With the prefix in place, every output sample of the block is a CIRCULAR convolution of the channel with the block:

$$\mathbf{y} = \mathbf{h} \circledast_N \mathbf{x}$$

This is the crucial step, because circular convolution in time becomes plain MULTIPLICATION under the DFT:

$$Y(k) = H(k)X(k), \quad 0 \leq k \leq N - 1$$

where $H(k)$ is the DFT of the channel taps, zero-padded to length N :

$$H(k) = \text{DFT} \left\{ h(0), h(1), \dots, h(L-1), \underbrace{0, \dots, 0}_{N-L} \right\}$$

THE PAYOFF

A messy linear convolution, which mixed symbols together, has been converted into a circular one, and the DFT turns that into an independent scalar product on every subcarrier. This is the mathematical heart of OFDM.

Transmitter and Receiver with Cyclic Prefix

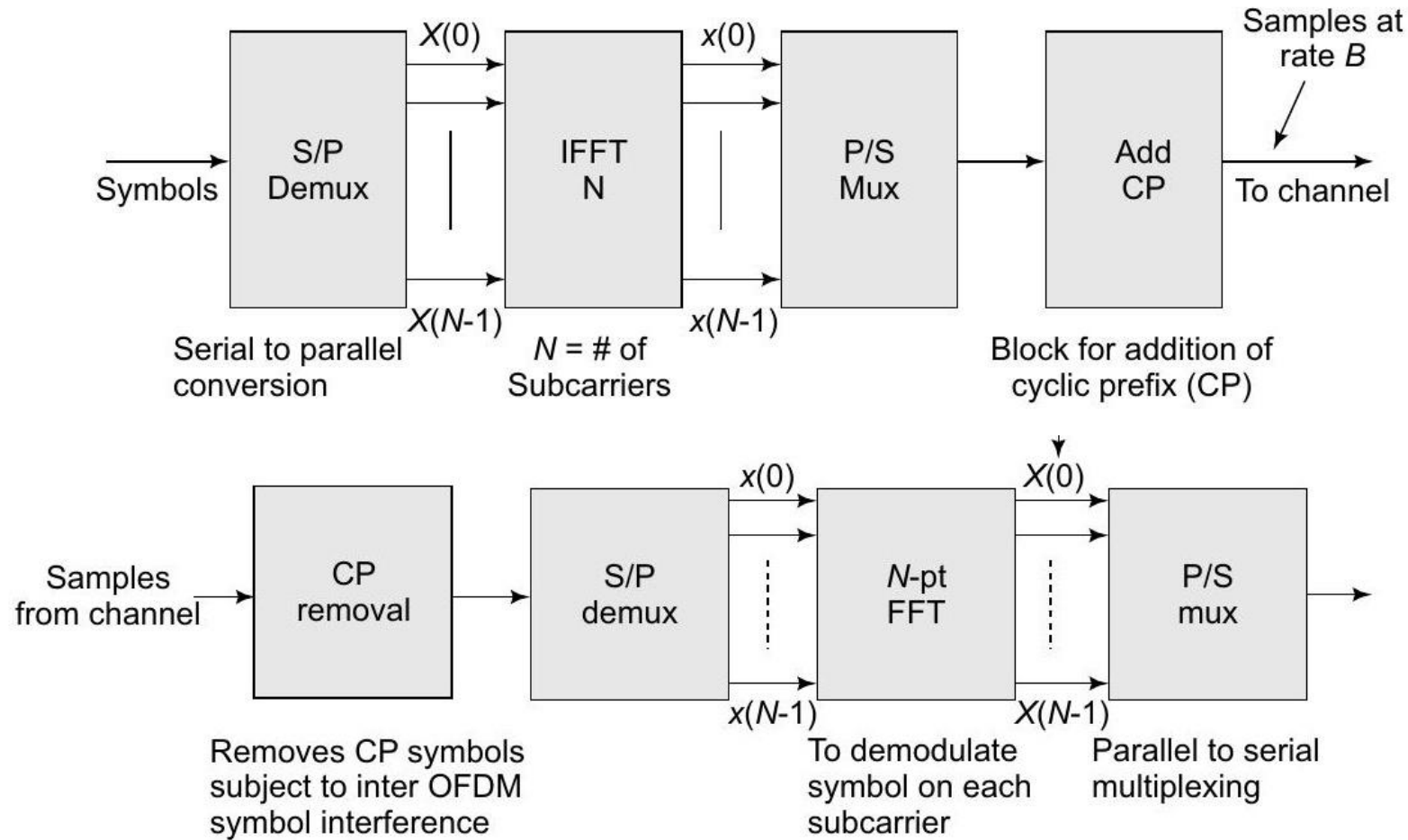


Figure 7.9 OFDM receiver schematic with CP

Figure 7.9: OFDM receiver schematic with CP (Jagannatham, 2016)

N Parallel Flat-Fading Subchannels

Including receiver noise, each subcarrier obeys its own independent flat-fading model:

$$Y(k) = H(k)X(k) + N(k)$$

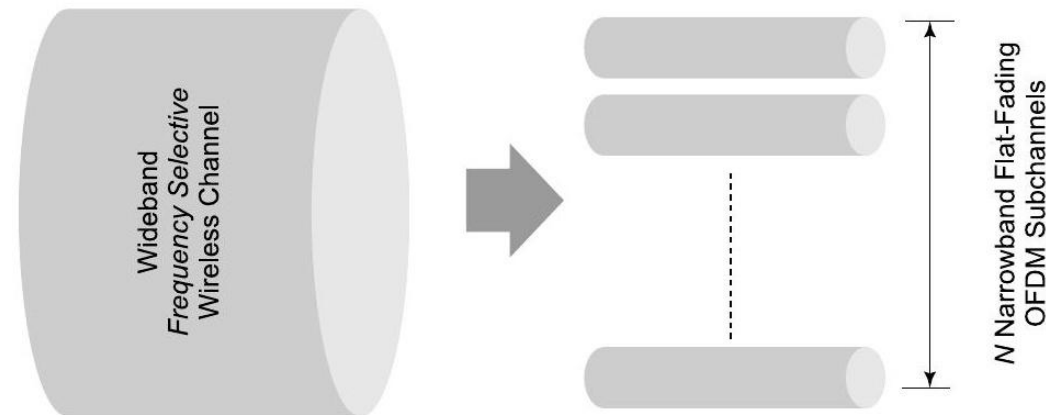


Figure 7.7 OFDM parallel subchannels

Figure 7.7: OFDM parallel subchannels (Jagannatham, 2016)

This is exactly the flat-fading model of Lecture 2, now holding on every subcarrier of a wideband system. Everything derived in Lectures 3 to 5 applies directly, per subcarrier.

BER of OFDM: Noise and Channel Statistics

The FFT output noise on subcarrier k is a sum of N time-domain noise samples:

$$N(k) = \sum_{m=0}^{N-1} n(m) e^{-j2\pi \frac{km}{N}}$$

Its power follows from whiteness: the double sum collapses to N diagonal terms:

$$\sigma_N^2 = E[|N(k)|^2] = \sum_{m=0}^{N-1} \sum_{l=0}^{N-1} E[n(m)n^*(l)] e^{-j2\pi(m-l)\frac{k}{N}} = N\sigma_n^2$$

The channel gain on each subcarrier collects the power of all L taps:

$$E[|H(k)|^2] = \sum_{m=0}^{L-1} E[|h(m)|^2] = L$$

Taking the ratio gives the average SNR per subcarrier:

$$\overline{\text{SNR}} = \frac{LP}{N\sigma_n^2}$$

Note the factor L : OFDM inherently collects the energy of ALL multipath taps on every subcarrier.

BER of OFDM: The Result

Each subcarrier is a flat Rayleigh channel, so the BER expression derived in Lecture 3 applies directly with this SNR substituted:

$$\text{BER}_{\text{OFDM}} = \frac{1}{2} \left(1 - \sqrt{\frac{LP/(N\sigma_n^2)}{2 + LP/(N\sigma_n^2)}} \right)$$

- Each subcarrier experiences flat Rayleigh fading, so its BER decays only as $1/\text{SNR}$: diversity order 1.
- Without coding, OFDM does NOT provide diversity by itself. It converts a hard equalization problem into many easy detection problems, but each subcarrier remains vulnerable.
- Coding across subcarriers is what recovers the frequency diversity latent in the L multipath taps, which is why coded OFDM is universal in practice.

The honest summary: OFDM solves ISI, not fading. Diversity still has to be bought separately, through coding, multiple antennas, or both.

Per-Subcarrier Detection

Because each subcarrier is a scalar channel, detection is a ONE-TAP operation. The zero-forcing detector divides out the channel:

$$\hat{X}(k) = \frac{1}{H(k)} Y(k) = X(k) + \frac{N(k)}{H(k)}$$

$$\tilde{N}(k)$$

The matched filter alternative avoids the division entirely:

$$H^*(k)Y(k) = |H(k)|^2 X(k) + H^*(k)N(k)$$

and the MMSE detector of Lecture 7 balances the two:

$$\hat{X}_{MMSE}(k) = \frac{H^*(k)}{|H(k)|^2 + \sigma_n^2} Y(k)$$

THE COST OF ZERO-FORCING

The noise term $N(k)/H(k)$ explodes on any subcarrier in a deep fade, exactly the noise amplification seen in Lecture 7. Some subcarriers will always be badly faded, which is why practical systems are CODED OFDM: coding across subcarriers rescues the faded ones using the strong ones.

The Price of the Cyclic Prefix

The prefix carries no new information, so it costs data rate. Sending $N + L_c$ samples to convey N symbols:

$$\text{Loss in efficiency} = \frac{L_c}{N + L_c} = \frac{L - 1}{N + L - 1} \xrightarrow{N \rightarrow \infty} 0$$

How long must the prefix be? It must span the channel delay spread T_d :

$$L_c \times T_s \geq T_d \Rightarrow L_c \geq \frac{T_d}{T_s} = \frac{B}{B_c}$$

Combining the two requirements gives a single elegant chain:

$$N \gg L_c \geq \frac{B}{B_c} \Leftrightarrow B_c \gg \frac{B}{N} \quad (\text{the flat-fading condition again})$$

EVERYTHING CLOSES THE LOOP

The condition for negligible prefix overhead is EXACTLY the condition that each subcarrier be flat-fading, $B/N \ll B_c$. The same inequality that made the subcarriers narrowband also makes the cyclic prefix cheap. One design choice, two benefits.

Worked Example: WiMAX OFDM Parameters

WiMAX uses $N = 256$ subcarriers with 15.625 kHz spacing and a cyclic prefix of 12.5%. Find the bandwidth, symbol duration, prefix length and rate loss.

Step 1: total signal bandwidth.

$$\frac{B}{N} = 15.625 \text{ kHz} \Rightarrow B = 256 \times 15.625 \text{ kHz} = 4 \text{ MHz}$$

Step 2: OFDM symbol duration before the prefix.

$$\frac{N}{B} = \frac{256}{4 \times 10^6} = 64 \mu\text{s}$$

Step 3: prefix duration and total symbol time.

$$\text{CP duration} = 12.5\% \times 64 \mu\text{s} = 8 \mu\text{s} \Rightarrow \text{total} = 64 + 8 = 72 \mu\text{s}$$

Step 4: prefix length in samples.

$$L_c = \frac{8 \mu\text{s}}{1/B} = 8 \times 10^{-6} \times 4 \times 10^6 = 32 \text{ samples}$$

Step 5: resulting rate loss.

$$\text{Loss} = \frac{32}{288} = \frac{8 \mu\text{s}}{72 \mu\text{s}} = 11.1\%$$

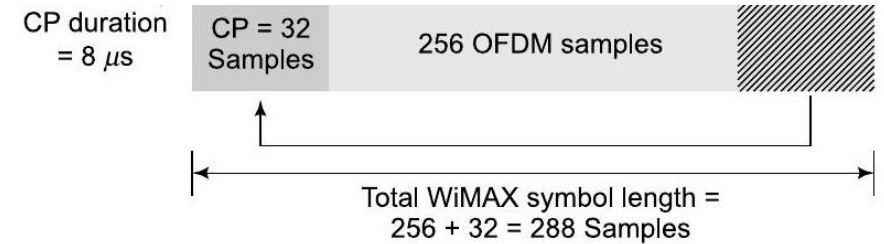


Figure 7.11 WiMAX OFDM symbol with cyclic prefix

ANSWERS

B = 4 MHz
T = 64 μs
CP = 8 μs
 $L_c = 32$ samples
Total = 288
Loss = 11.1%

MIMO-OFDM

Combine the two most important technologies in modern wireless. With multiple antennas the channel becomes a matrix FIR filter:

$$\mathbf{y}(n) = \sum_{l=0}^{L-1} \mathbf{H}(l) \mathbf{x}(n-l) + \mathbf{w}(n)$$

Applying the OFDM machinery on EACH transmit and receive antenna decouples this into N independent MIMO systems, one per subcarrier:

$$\tilde{\mathbf{y}}(k) = \tilde{\mathbf{H}}(k) \tilde{\mathbf{x}}(k), \quad 0 \leq k \leq N-1$$

$$\hat{\tilde{\mathbf{x}}}_{ZF}(k) = \left(\tilde{\mathbf{H}}^H(k) \tilde{\mathbf{H}}(k) \right)^{-1} \tilde{\mathbf{H}}^H(k) \tilde{\mathbf{y}}(k)$$

$$\hat{\tilde{\mathbf{x}}}_{MMSE}(k) = P_d \left(P_d \tilde{\mathbf{H}}^H(k) \tilde{\mathbf{H}}(k) + \sigma_w^2 I \right)^{-1} \tilde{\mathbf{H}}^H(k) \tilde{\mathbf{y}}(k)$$

MIMO-OFDM is the physical layer of LTE, LTE-Advanced, 802.11n/ac/ax and 5G NR.

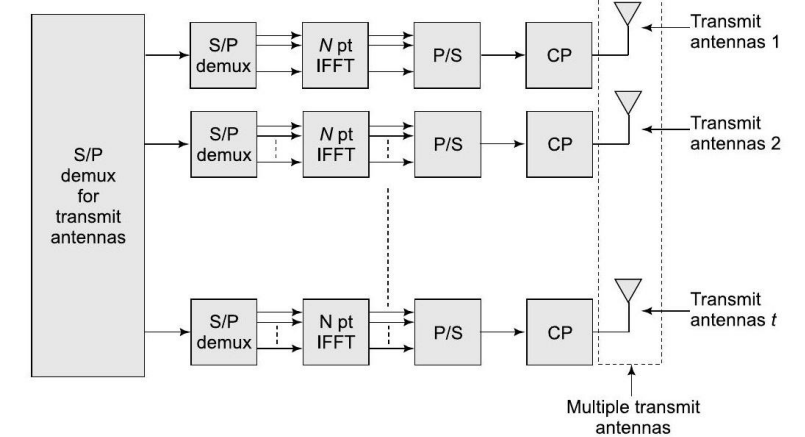


Figure 7.12 MIMO OFDM transmitter schematic

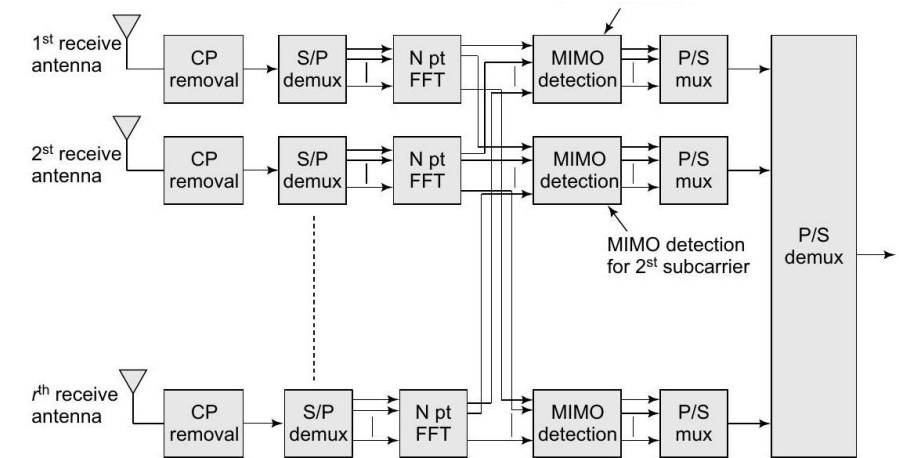


Figure 7.13 MIMO-OFDM receiver schematic

Carrier Frequency Offset: The Problem

OFDM rests entirely on subcarrier orthogonality, which in turn rests on the receiver oscillator being exactly aligned with the transmitter's. Any mismatch Δf destroys that alignment.

Normalize the offset to the subcarrier spacing:

$$\epsilon = \frac{\Delta f}{B/N} \quad (\text{offset normalized to subcarrier spacing})$$

The received samples now carry an extra rotating phase:

$$y(n) = \frac{1}{N} \sum_{k=-N/2}^{N/2} X(k)H(k)e^{j2\pi n \frac{k+\epsilon}{N}} + w(n)$$

With no offset, the orthogonality sum vanishes for $k \neq l$ and each subcarrier separates cleanly:

$$\epsilon = 0: \sum_n e^{j2\pi n \frac{k-l}{N}} = 0 \quad (k \neq l) \Rightarrow Y(l) = X(l)H(l) + W(l)$$

BUT WITH $\epsilon \neq 0$

The exponent is no longer an integer multiple of $2\pi/N$, so the sum does NOT vanish. Every subcarrier leaks into every other one. This leakage is called Inter-Carrier Interference (ICI), and it is the single greatest practical vulnerability of OFDM.

Deriving the ICI Power

Using the geometric sum identity for the phase terms:

$$\sum_{n=0}^{N-1} e^{j\theta n} = \frac{\sin(N\theta/2)}{\sin(\theta/2)} e^{j\tilde{\phi}}$$

the received symbol on subcarrier l splits into a scaled desired term plus leakage from every other subcarrier:

$$Y(l) = \underbrace{H(l)X(l) \frac{\sin(\pi\epsilon)}{N\sin(\pi\epsilon/N)} e^{j\tilde{\phi}_l}}_{\text{desired}} + \sum_{k \neq l} \underbrace{H(k)X(k) \frac{\sin(\pi\epsilon)}{N\sin\left(\pi \frac{l-k+\epsilon}{N}\right)} e^{j\tilde{\phi}_{kl}}}_{\text{ICI}} + W(l)$$

For large N the desired scaling simplifies to a sinc, giving the attenuated signal power:

$$N\sin\left(\frac{\pi\epsilon}{N}\right) \rightarrow \pi\epsilon \Rightarrow \text{Signal power} = P\overline{|H|^2} \left(\frac{\sin(\pi\epsilon)}{\pi\epsilon}\right)^2$$

Bounding the interference uses the inequality $\sin(\pi u/N) \geq 2u/N$, then the Basel sum:

$$\sin\left(\frac{\pi u}{N}\right) \geq \frac{2u}{N} \Rightarrow N\sin\left(\frac{\pi u}{N}\right) \geq 2u \quad E[|I_l|^2] \leq \frac{1}{2} P\overline{|H|^2} \sin^2(\pi\epsilon) \sum_{u=1}^{\infty} \frac{1}{u^2} = \frac{\pi^2}{12} P\overline{|H|^2} \sin^2(\pi\epsilon) = 0.822 P\overline{|H|^2} \sin^2(\pi\epsilon)$$

$\pi^2/6$

The constant 0.822 comes from $\pi^2/12$, which is where the Basel sum $\sum 1/u^2 = \pi^2/6$ enters.

Example: SINR Loss from a Frequency Offset

Combining signal and interference gives the SINR under a carrier frequency offset:

$$\text{SINR} = \frac{P\overline{|H|^2} \left(\frac{\sin \pi \epsilon}{\pi \epsilon} \right)^2}{0.822 P\overline{|H|^2} \sin^2(\pi \epsilon) + \sigma_n^2}$$

A WiMAX system at SNR = 10 dB has $\epsilon = 0.05$. Find the SINR, and the oscillator accuracy needed at 2.4 GHz.

Step 1: with no offset the SNR is simply:

$$\epsilon = 0: \quad \text{SNR} = \frac{P\overline{|H|^2}}{\sigma_n^2} = 10 \text{ dB} = 10$$

Step 2: substitute $\epsilon = 0.05$ into the SINR expression.

$$\epsilon = 0.05: \quad \text{SINR} = \frac{10 \left(\frac{\sin \pi (0.05)}{\pi (0.05)} \right)^2}{0.822 \times 10 \sin^2(\pi \times 0.05) + 1} = 8.25$$

Step 3: convert the normalized offset into an absolute frequency and an oscillator tolerance.

$$\Delta f = \epsilon B_s = 0.05 \times 15.625 \text{ kHz} \approx 0.78 \text{ kHz}$$

$$\frac{0.78 \times 10^3}{2.4 \times 10^9} = 1/3 \times 10^{-6} = 0.33 \text{ ppm}$$

A mere 5% offset drops the SINR from 10 to 8.25, and demands oscillator accuracy of 0.33 ppm. OFDM is extremely sensitive to frequency synchronization.

Peak-to-Average Power Ratio: Deriving PAPR = N

PAPR measures how much the signal envelope fluctuates:

$$\text{PAPR} = \frac{\text{Peak power}}{\text{Average power}}$$

For a single carrier with constant-modulus symbols the envelope never varies:

$$\text{Single carrier: Peak} = \text{Average} = a^2 \Rightarrow \text{PAPR} = 1 = 0 \text{ dB}$$

For OFDM the time samples are the IFFT of the symbols:

$$x(k) = \frac{1}{N} \sum_{i=0}^{N-1} X(i) e^{j2\pi \frac{ki}{N}}$$

Average power, using that the symbols are uncorrelated:

$$E[|x(k)|^2] = \frac{1}{N^2} \sum_{i=0}^{N-1} E[|X(i)|^2] = \frac{1}{N^2} \times a^2 N = \frac{a^2}{N}$$

Peak power, when all N symbols happen to align in phase:

$$|x(k)| \leq \frac{1}{N} \sum_{i=0}^{N-1} \underbrace{|X(i)|}_a \underbrace{|e^{j2\pi ki/N}|}_1 = a \Rightarrow \text{Peak power} = a^2$$

$$\text{PAPR}_{\text{OFDM}} = \frac{a^2}{a^2/N} = N$$

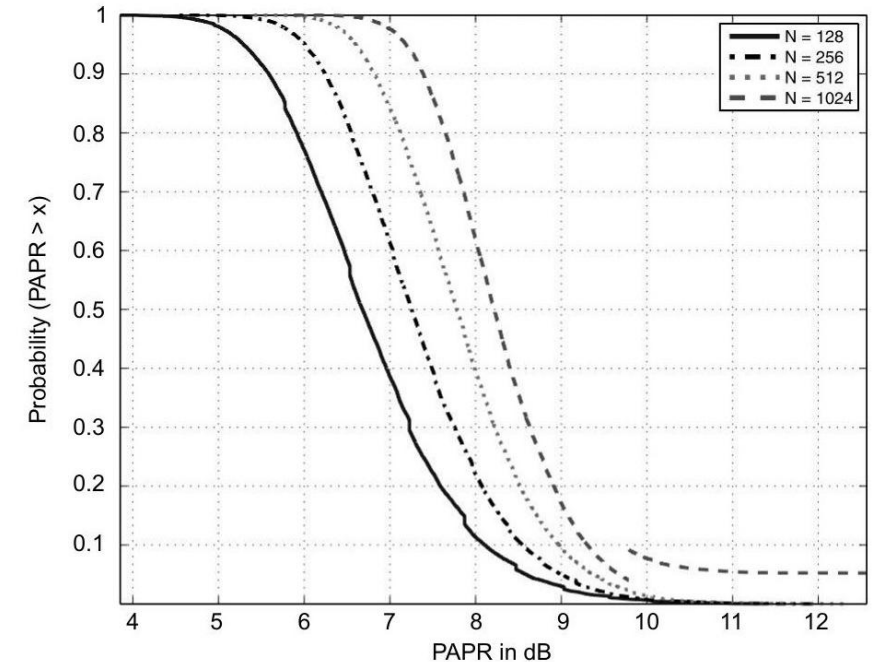


Figure 7.15 OFDM PAPR for various number of subcarriers N

Why High PAPR Hurts

A PAPR of N means that with $N = 256$ the peak power can be 256 times the average, roughly 24 dB. Two consequences follow:

- The power amplifier must stay LINEAR up to the peak, not the average. Amplifiers are linear only over a limited range, and saturate beyond it.
- Saturation clips the signal, producing nonlinear distortion that destroys subcarrier orthogonality and regrows spectrum into neighbouring bands.
- Avoiding this requires backing off the operating point, which wastes amplifier efficiency and drains battery in handsets.

PAPR is characterized statistically by the complementary CDF, the probability that it exceeds a threshold:

$$\bar{F}_X(x) = \Pr(X > x) = 1 - F_X(x)$$

This is why LTE uses OFDMA on the DOWNLINK but SC-FDMA on the UPLINK: the base station can afford a large linear amplifier, the handset cannot.

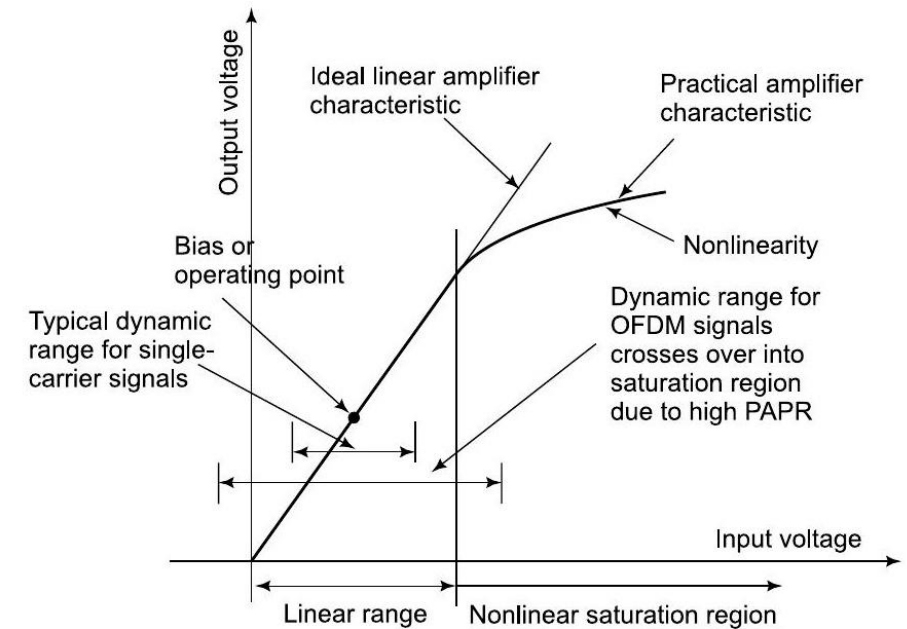


Figure 7.16 Nonlinear amplifier characteristic

SC-FDMA: Undoing the IFFT

Start from a hypothetical modification: place an N-point FFT immediately BEFORE the N-point IFFT in the OFDM transmitter.

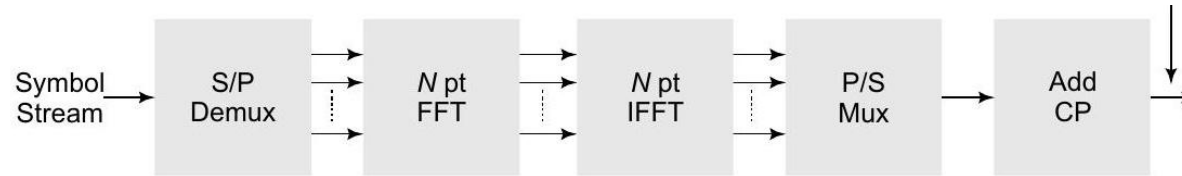


Figure 7.17 Hypothetical modification of OFDM transmitter

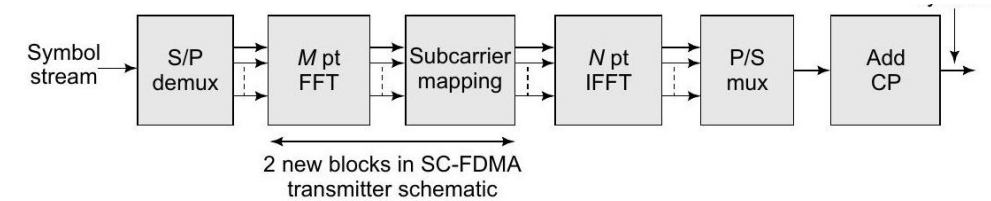


Figure 7.18 SC-FDMA transmitter schematic

$$N\text{-pt FFT} \rightarrow N\text{-pt IFFT} = \text{identity} \Rightarrow \text{PAPR} = 0 \text{ dB}$$

An FFT followed by an IFFT of the SAME size is the identity operation, so the transmitted signal is simply the original single-carrier stream, which has 0 dB PAPR.

The practical version uses a SMALLER M-point FFT, so the cancellation is only partial:

$$M < N: \quad M\text{-pt FFT} \rightarrow \text{mapping} \rightarrow N\text{-pt IFFT}$$

PAPR lands between single-carrier and full OFDM, much closer to single-carrier.

SC-FDMA Receiver and Subcarrier Mapping

The receiver mirrors the transmitter: after the N-point FFT and per-subcarrier equalization, an M-point IFFT undoes the transmitter's M-point FFT.

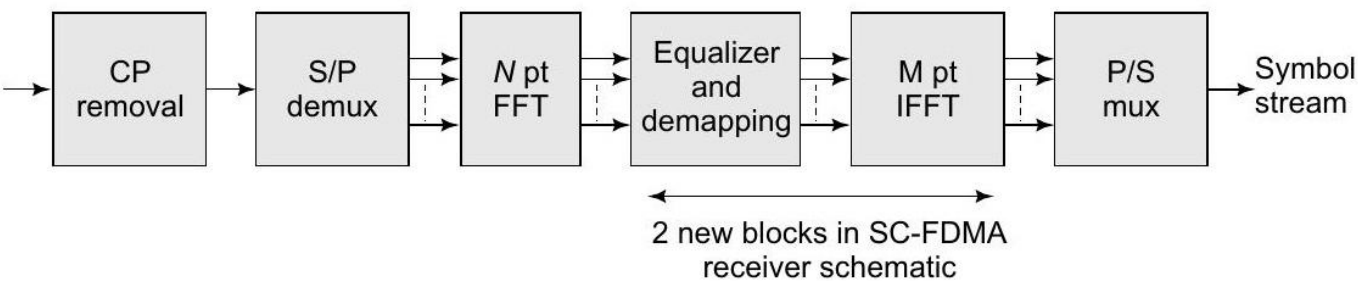
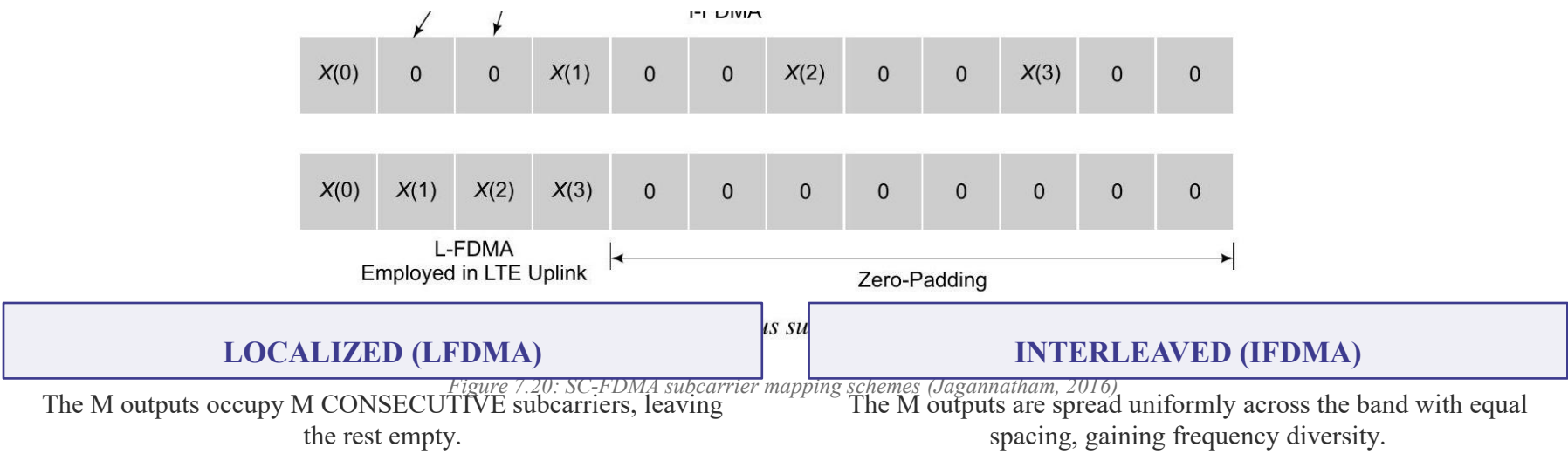


Figure 7.19 SC-FDMA receiver schematic

The M outputs of the transmitter FFT must be mapped onto N subcarriers. Two schemes are standard:



OFDMA and Subcarrier Loading

OFDMA extends OFDM to multiple users by allocating DIFFERENT subcarriers to different users, since the subcarriers are already orthogonal.

- Each user's channel is frequency-selective, so different users see peaks and nulls at DIFFERENT subcarriers.
- Assign each subcarrier to whichever user currently sees the best gain on it: this is multi-user diversity.
- Adaptive loading goes further: vary the modulation order per subcarrier, sending more bits where the channel is strong.

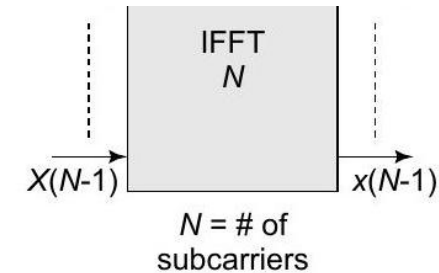


Figure 7.14 *OFDM subcarrier loading*

This is the same water-filling logic as MIMO in Lecture 7, applied across FREQUENCY rather than across spatial modes: give resources to the strong channels and starve the weak ones.

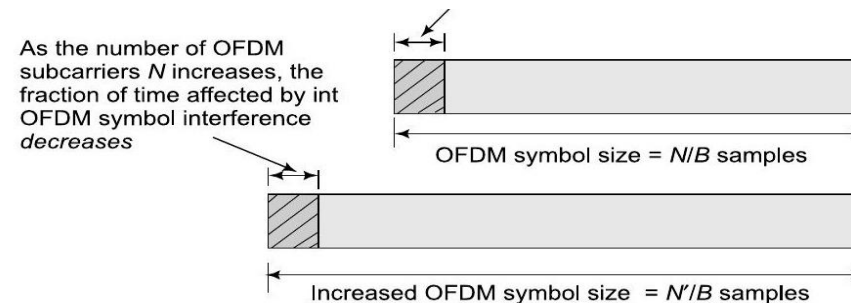


Figure 7.10 *Inter OFDM symbol interference with increasing OFDM symbol time*

Lecture 4: Key Takeaways

- A wideband single-carrier signal over a frequency-selective channel suffers ISI, and equalizer complexity grows with delay spread.
- Dividing the band into N subcarriers makes each narrowband ($B/N \ll B^c$) and therefore flat, at the SAME overall symbol rate B .
- Subcarriers are orthogonal because integrating a sinusoid at an integer multiple of the fundamental over one period gives exactly zero.
- The sampled multicarrier signal IS an IDFT of the data, so one IFFT replaces the impractical bank of N modulators.
- The cyclic prefix ($L^c \geq L-1$) removes inter-OFDM-symbol interference and converts linear convolution into circular, giving $Y(k) = H(k)X(k) + N(k)$.
- The condition for cheap cyclic prefix overhead is exactly the flat-fading condition: one design choice, two benefits.
- OFDM solves ISI but NOT fading: each subcarrier still decays as $1/\text{SNR}$, so coding across subcarriers is essential.
- Its two weaknesses are sensitivity to carrier frequency offset, which causes ICI, and a PAPR of N , which SC-FDMA addresses on the uplink.

OFDM underpins LTE, LTE-Advanced, WiMAX, 802.11a/g/n/ac/ax, DVB, DAB and 5G NR.

Lecture Reference

PRIMARY SOURCE

Aditya K. Jagannatham, Principles of Modern Wireless Communication Systems, McGraw Hill Education (India), 2016, Chapter 7 in full, Sections 7.1 to 7.8, including Example 7.1.

BOOK VISUALS USED

All twenty chapter figures, 7.1 through 7.20, used directly for classroom teaching, together with the WiMAX design example of Section 7.3.

NOTE ON THIS EDITION

This deck replaces the earlier Lecture 4, which covered only Sections 7.1 to 7.4 and deferred the rest. It now covers the complete chapter: OFDM BER (7.4), MIMO-OFDM (7.5), carrier frequency offset and ICI (7.6), PAPR (7.7), and SC-FDMA (7.8), with every derivation and figure included.